

MTH204 (Section A)
Graded Worksheet–2

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

1. For the differential equation

$$\frac{dx}{dt} = -x^3 + 5x^2 - 6x,$$

find all equilibrium points and discuss their nature.

2. **Which of the following statements is/are correct?** (One or more options may be correct. No partial marking.)

Consider the IVP

$$\frac{dy}{dx} = x y^{1/3}, \quad y(0) = 0, \quad (x, y) \in \mathbb{R}^2.$$

Determine which of the following statements are true:

- (a) The function $f(x, y) = xy^{1/3}$ does not satisfy a Lipschitz condition in y near $y = 0$.
 - (b) There is a unique solution to the IVP.
 - (c) The IVP has no solution.
 - (d) The IVP has more than one solution.
3. A model for the spread of contagious diseases assumes that the rate of spread is proportional to the number of contacts between infected and non-infected persons, who move freely among one another. Set up the model. Find the equilibrium solutions and determine their stability. Solve the ODE and find the limiting proportion of infected persons as $t \rightarrow \infty$, explaining its meaning.

4. The lines of electric force of two opposite charges located at $(-1, 0)$ and $(1, 0)$ are the family of circles

$$x^2 + (y - c)^2 = 1 + c^2.$$

Show that the equipotential lines (orthogonal trajectories) are the circles

$$(x - k)^2 + y^2 = k^2 - 1.$$

5. Solve the differential equation

$$y' + x^2 y = \frac{e^{-x^3} \sinh(x)}{3y^2}.$$

Solution:

Solution:

Solution: